Benchmarking Counterfactual Explanations of Model Change with Exact Ground Truth

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Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method sees which change. We contribute a benchmark with exact ground truth and a linear analysis of what counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE: nearest inputs on which exactly one model changes its decision by a margin) can see. Used as a probe, Δ-CE abstains on null updates of CNNs (0 of 2 400 unfiltered queries, and 0 of 800 when it searches through a learned VAE decoder; both registered), while differencing, IG- and SHAP-differences report a change; on a ResNet-8 the margin must first be calibrated on validation nulls (exploratory). As predicted, gradient differences are blind to an offset and see a rotation; a pre-registered hypothesis that Δ-CE recovers the changed factor better than they do was rejected. Whether decisions moved can be read from a margin score; the witness adds a verified input showing where.

1 Introduction

An update that improves accuracy can still change individual decisions in unwanted ways [11]; explaining *how* they differ is a goal of explainable AI under distributional change [3, 12], distinct from explaining a shift of the data [14]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed. Benchmarks with ground truth exist for explaining one model [15, 16, 17], not for explaining a change, so this goes unchecked.

We ask which kind of change each family can see. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one. We use it as a probe that should stay silent when nothing changed and, when it answers, return an input that can be checked.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a 4×4 grid over the rotation and threshold of the decision boundary, with code and protocol. (ii) A linear analysis separating rotation from offset, with a prediction for each family. (iii) Tests on CNNs, a ResNet-8 and a learned latent space under a protocol fixed in advance, including follow-ups registered before their data existed; all failures are reported. The central result: a margin witness stays silent on null updates, where differencing and attribution differences report change.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness. *Proof:* subtract the two constraints of $\mathcal{F}_\kappa^N$.

Corollary 1 (constant logit offset). If f_{new} adds b to the logit of class 1, e.g. the prior correction [13], then $h \equiv \pm b$ and $\nabla h \equiv 0$. Type-N witnesses exist only for the class whose margin b lowers and only if $|b| \geq 2\kappa$, at the points with $\kappa \leq m_o \leq |b| - \kappa$. *Proof:* insert $h \equiv -|b|$ in Lemma 1; no linearity is needed.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing $u_n - u_o$ (the two minimum-norm counterfactuals), the witness $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference lies mainly along the new dependence v; the witness is dominated by v iff $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$; after projecting out $\hat{w}_o$ all three coincide up to scale. *Proof* of (a): both normals equal $\hat{w}_o$; of (b): supplement.

In words, gradient differences see a *rotation* and are blind to an *offset*; type-N witnesses exist only on the side whose margin the offset lowers. The analysis was formulated after the main experiment and before the grid was run.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. All methods search the same factor space through the same renderer, so no method has access the others lack. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data. For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions and

Table 1: Injected changes; $k^\star$: ground-truth factor. [a]$\alpha \in \{0, .1, .25, .5\}$, $t \in \{.5, .53, .56, .6\}$, two seeds; four offsets enter the tests. [b]Split D, registered; 8 qualified.

Change	Adaptation data	$k^\star$	Pairs
Shortcut	colour correlated with label (0.99)	colour	3
Subgroup	labels flipped for size > 0.75	size	3
Threshold	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid[a]	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$	aspect / colour	8 / 24
Offsets[b]	grid, $\alpha = 0$, $t \in \{.53, .56, .6\}$, seeds 2–4	aspect	9

analyse at these values.

Metrics and methods. Per pair: $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0). Recovery uses type-N witnesses only (pre-declared); a query whose class the update raises (Threshold, class 0) has none and counts 0. All counterfactual methods share one solver (Adam, then SLSQP), the points and the budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1]; **Grad-diff**, the gradient difference [1]; **Imp-diff**, a difference of permutation importances on the test data [4], one vector per pair; **Seg-search**, a line search along segments from the query to training points, after [5]; **No-prox**, Δ-CE without the proximity term (the DeepXplore objective [6]); and **CF-new**, the counterfactual of f_{new} alone. DiG-IN [7] searches a learned generator for disagreement; on an exact manifold its objective reduces to No-prox. Section 4.4 adds IG- [18] and SHAP-differences [19].

Protocol. Every test split served at most one confirmatory test, after all choices were fixed: split A is the locked split of Table 2, split B the new data of the tests formed after the first results, split C the data of the type-O analysis, split D the data of three follow-ups registered [20] after the earlier results and before split D was generated, split F that of the latent-space test [21]. Bootstraps resample points within pairs, so their intervals do not cover model-to-model variation; Table 3 therefore also counts pairs. The rest is descriptive. A digit tier and a label-shift update failed qualification.

4 Results

4.1 Abstention and detection

No disagreement search returns a point on a null update (Table 2); on the 8 null pairs and on five null updates of an MLP on Adult no witness exists, while CF-diff attributes a change to 97.7–100% of points. As these nulls were admitted only without decisive disagreement on validation, close to the witness criterion, we registered 8 new null pairs admitted without any check (split D): again no witness on any of 2 400 queries, CF-diff attributes a change to 98.3–99.7%. Nor

Table 2: Recovery@r (found-rate) with type-N Δ-CE, split A (locked), mean over three pairs (95% CIs within ± 0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	Seg-search	No-prox	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

Table 3: Tests, grouped as in Section 4 (95% CIs; †90%). Pairs: pairs on the claimed side / pairs. expl. = exploratory. K: D = follows from a lemma, proposition or definition; R = risky prediction. When: pre = before any result; post = after the first, on split B; grid = before the grid run; new = later, new data; reg = registered [20]. Holm families: *, ‡, §. ¶Run after its registered cost rule had excluded it. Proj.: $\hat{w}_o$ projected out. $^{\|}$Point level only: with ≤ 4 pairs the exact pair-level $p \geq 0.0625$; the other confirmed rows also hold with the pair as unit (supplement).

Claim	Data	Estimate	p	Pairs	Status	K, when
Abstention						
No witness on nulls	Null, B; Adult	0/2 400; 0/1 500		8/8; 5/5	confirmed	D, post; new
No witness on unfiltered nulls	Null, D	0/2 400		8/8	confirmed	R, reg
Same, search via VAE decoder	Null, F	0/800		8/8	confirmed	R, reg
No witness on nulls	ResNet-8, D¶; E, $\kappa^{\star}$	180/200; 0/200		0/2; 2/2	not; expl.	R, reg; new
CF-diff (no cut) flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$ §		confirmed	D, new
Offsets						
Grad-diff < 0.2	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$ ‡	4/4	confirmed$^{\|}$	R, grid
Grad-diff < 0.2	Offsets, D	0.033–0.093	0.004 (sign)	8/8	confirmed	R, reg
Grad-diff offset-free	Grid	-0.098 [-0.143, -0.053]†	0.94 ‡		not conf.	R, grid
Δ-CE $>$ Grad-diff	Threshold, B	$+0.31$ [0.27, 0.35]	$2{\cdot}10^{-4}$ *	3/3	confirmed$^{\|}$	D, post
Rotations						
Grad-diff rises with α	Grid	$+0.57$ [0.53, 0.61]	$4{\cdot}10^{-4}$ ‡	$\rho = 0.89$	confirmed	R, grid
Proj. Δ-CE $\approx$ Grad-diff	Shortc. B; grid	-0.30; -0.35 (CIs ± 0.03)†	1 *; 1 ‡		not conf.	R, post; grid
Normal at $z^{\star}$ halves gap or more	Grid, new	$-0.19 \to +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$ §	4/4	confirmed$^{\|}$	R, new
Δ-CE versus gradient differences						
Δ-CE $>$ Grad-diff	Subgroup	-0.09 [-0.13, -0.05]		0/3	rejected	R, pre
Tie ($\lvert\Delta\rvert \leq 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$			not conf.	R, pre

does silence need the exact renderer: searching through the decoder of a VAE trained on the images (registered [21]; 100 queries per pair, on CPU instead of the registered GPU), no null query has a witness (0/800; CF-diff 85–94%), while on 6 of 8 offsets witnesses exist for at least half of the lowered-class queries (CF-diff none at zero false positives). The margin is what makes the probe silent: with $\kappa = 0$ witnesses appear on 83% of the split-A null queries. On the offsets the witness rate rises from 0.02 to 1.00 with the largest $|h|$ in the r-ball (descriptive): the probe reads decisive change, in units of κ.

Does abstention cost sensitivity? We cut each score on the nulls to flag none (or 5%) and count flagged lowered-class queries. For a constant logit offset b, witnesses appear on every lowered-class query from $|b| = 1.1$ (Corollary 1; below 2κ they miss real changes by design), CF-diff flags ≤ 0.002 up to $|b| = 2$, and the margin score $|h(z_0)|$ flags all from $|b| = 0.9$ at 5%. On the 8 registered fine-tuned offsets, at zero false positives, witnesses exist for 0.17–1.00 of lowered-class queries, the margin score flags 0.00–0.54 and CF-diff none (0.03–0.18 at 5%). The witness exceeded the margin score by 0.2 in 7 of 8 pairs, not in all as registered, so we

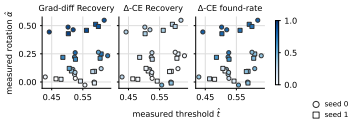


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (grid test split), on one colour scale from 0 to 1. Hollow: no $k^\star$; marker shape: seed of f_{old}.

claim no advantage in telling *whether* decisions moved; a margin score does that cheaply. What the witness adds is *where*: a verified input near the query.

4.2 Offsets

Gradient differences are blind to an offset, as Proposition 2(a) predicts: on the pure offsets of the grid their Recovery is 0.081 (below 0.2 in each of the 4 pairs), and on the 8 registered offsets of split D 0.03–0.09 (8 of 8 below 0.2; sign test $p = 0.004$). On Threshold Δ-CE beats them (0.41 vs. 0.10 on split B; positive for each seed). Type-N witnesses exist only on the side the offset lowers (Threshold: class 1), hence 0.47 in Table 2; type O covers the other side (split C).

4.3 Rotations

Gradient differences see a rotation, increasingly with its angle: Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ (Spearman $\rho = 0.89$ over the 24 pairs, permutation $p = 10^{-4}$; Fig. 1). The projection equivalence of Proposition 2(b) holds only with $\hat{w}_o$ taken at the witness (Table 3).

4.4 Other baselines and a second architecture

With predictions logged beforehand (split A), IG-diff and SHAP-diff recover colour on Shortcut (0.93) like Grad-diff, and name aspect on Threshold (0.73–0.75 vs. 0.07), against the predicted blindness. The nulls show that this is no signal: there IG- and SHAP-diff name aspect on 57% and 71% of queries (Grad-diff 5%), and IG-diff on 42% of the split-D nulls, as predicted. Imp-diff on test data ranks aspect first in all 9 change pairs (Table 2), and $k^\star$ on adaptation data. With ten times the segments Seg-search finds witnesses for 0.26–0.47 of queries (predicted ≤ 0.2); on all 955 of them Δ-CE finds one too, nearer or equal on 953.

A registered ResNet-8 replication (split D, 100 queries per pair; run after its cost rule had excluded it) shows that κ is model-specific: null fine-tuning moves its margins four times more and witnesses appear on 0.90 of null queries. Calibrating κ on validation nulls (rule fixed in advance, 0.5 for the CNN; grid extended after $\kappa \leq 2$ failed) gives $\kappa^\star = 2.5$: on split E no null query has a witness (0/200), and witnesses exist for 0.78–1.00 of lowered-class offset queries (margin score ≤ 0.05, CF-diff ≤ 0.03 at zero false positives). Recovery stays 0.36–0.40; Grad-diff is not offset-blind there (0.32–0.36).

5 Discussion and limitations

For *where* we have checks, not a score: witnesses are verified by definition, and the search misses few and returns nearer ones than No-prox and Seg-search. **Limitations.** (1) The data are synthetic and binary; through a learned decoder only abstention and detection were tested (*where* is then read approximately); multi-class tasks, other domains and real shifts are untested, and the tabular test covers abstention only. (2) κ is in logit units, exact only for a constant offset, and must be calibrated on null updates of each model family (0.5 for the CNNs; 2.5 for the ResNet-8, after an extended search). (3) Witnesses see only decision changes on the manifold within the budget, without a certificate; Recovery scores one notion of change, with one changed factor (smaller budgets and rank metrics: supplement). (4) Several tests were formed after earlier results; changes were logged before their data existed, with two disclosed exceptions; log, proofs, pair-level tests, code: [repository link].

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